



Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

AY: 2025-26

Class:	BE	Semester:	VIII
Course Code:		Course Name:	AAZ

Name of Student:	DART ANKIT VINOD
Roll No. :	61
Assignment No.:	2
Title of Assignment:	
Date of Submission:	
Date of Correction:	

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Completeness	5	3
Demonstrated Knowledge	3	2
Legibility	2	2
Total	10	07

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Completeness	5	3-4	1-2
Demonstrated Knowledge	3	2	1
Legibility	2	1	0

Checked by

Name of Faculty : Dr. Vaishali Shirsat

Signature :

Date :

17/13

2. 1) draw and explain architecture of GAN.

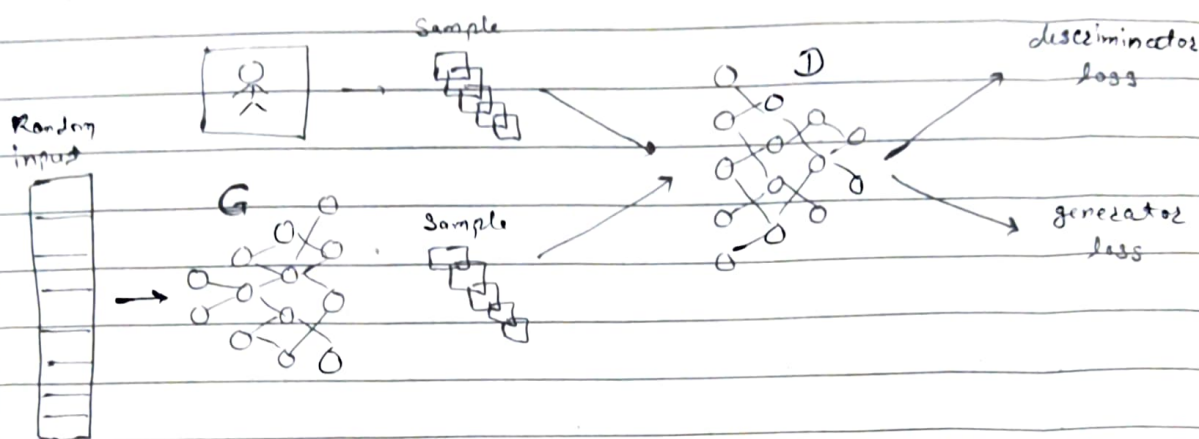
→ Architecture of GAN -

A GAN consist of two neural networks competing with each other :

(i) Generator (G)

(ii) Discriminator (D)

They are trained together in a minimax game.



GAN Architecture

Components of GAN -

(i) Generator (G) -

Input : Random noise vector z

Output : fake data sample (image, text etc)

goal : fool the discriminator.

(ii) discriminator -

Input - Real data + generated data

output - probability (Real = 1, fake = 0)

goal - correctly classify real vs fake

Mathematical objective function -

$$\min_G \max_D V(D, G) = E[\log D(x)] + E[\log (1 - D(G(z)))]$$

• meaning :

• discriminator maximizes -

- $\log D(x)$ $\rightarrow$ classify real correctly

- $\log (1 - D(G(z)))$ $\rightarrow$ classify fake correctly

• generator minimizes -

- $\log (1 - D(G(z)))$ $\rightarrow$ tries to fool D

• Training process -

i) generate fake data using generator.

ii) Train discriminator on :

- real data (label = 1)

- fake data (label = 0)

iii) Train generator - Try to make D classify fake as real

iv) repeat until equilibrium.

• Intuition :

• Think of it like,

- Generator - counterfeit money maker

- Discriminator - Police

Generator tries to make better fake money, police tries to detect fake money. over time, both improves.

• Types of GAN.

(i) DCGAN (deep convolutional GAN)

(ii) CGAN (conditional GAN)

(iii) WGAN (Wasserstein GAN)

(iv) Style GAN (high-quality face generation)

(v) Cycle GAN (Image-to-image translation)

2) To increase the stability of GAN, differentiate betⁿ the architectures of DCGAN and WGAN, which would be more suited, justifying your answer.

→ (i) DCGAN (Deep Convolutional GAN)

Architecture characteristics -

- Uses convolutional layers
- Uses batch normalization
- Uses ReLU (generator) and leaky ReLU (discriminator).
- final layer uses sigmoid.
- loss function :- binary cross entropy (BCE)
- Objective function -

$$\min_{\theta, \phi} \max_{\theta, \phi} = E[\log D(x)] + E[\log (1 - D(G(z)))]$$

- key properties -
- discriminator outputs probability (0-1)
- uses Jensen-Shannon divergence.
- Suffer from - mode collapse
 - vanishing gradients
 - training instability
 - sensitive hyperparameters.

(ii) WGAN - (Wasserstein GAN) -

- discriminator replaced by critic
- No sigmoid activation at output
- No probability output
- Uses weight clipping
- loss based on Wasserstein distance.

Objective function -

$$\min_{\theta} \max_{\phi} = E[D(x)] - E(D(G(z)))$$

Q. 3) To solve the problem of image reconstruction, differentiate between the architectures of conditional GAN and cycle GAN, which could be more suited, justifying your answer.

→ (i) CGAN (Conditional GAN).

- One generator (G)
- One discriminator (D)
- Both are conditioned on extra information
- The condition can be - class label
- another image

Objective function -

$$\min_G \max_D = E[\log D(x|z)] + E[\log (1 - D(G(z|z)))]$$

- requires paired dataset
- works well for, image super-resolution, image colorization, denoising, paired image-to-image translation.

(ii) CGAN - (Cycle GAN).

- Two generators - $G: X \rightarrow Y$ & $F: Y \rightarrow X$
- Two discriminators - D_x, D_y
- cycle consistency loss -

$$L_{\text{cycle}} = E[||F(G(x)) - x||] + E[||G(F(y)) - y||]$$

- After translation and reverse translation, original image is recovered.

- Does not require paired dataset
- learns mapping between two domains.
- Used for - style transfer
 - domain adaptation
 - unpaired image translation